

Craig Waite, Ph.D.

Assistant Professor of Chemistry

✉ waittc@trine.edu

id 0000-0003-4189-4103

☎ 260-665-4581

🌐 cwaite

📄 Best Hall 373, Trine University

🌐 <https://cwaite.github.io/>

Academic Appointments

Aug 2024 – Present | **Assistant Professor of Chemistry, Trine University**
Department of Chemical and Forensic Sciences, Angola, Indiana

Education

Jan 2018 – Jan 2023 | **University of Notre Dame**, Notre Dame, IN
Ph.D. in Chemistry
Dissertation: First-Principles Studies of Heterogeneous Catalytic Systems: Reactivity and Stability of Surfaces and Zeolites
Advisor: William F. Schneider, Ph.D.

Sep 2015 – May 2017 | **Montclair State University**, Montclair, NJ
M.S. in Chemistry
Thesis: A Computational Study of C–CN Bond Activation Through Nickel Catalysis Using the Random Phase Approximation
Advisor: Henk Eshuis, Ph.D.

Sep 2010 – May 2015 | **Montclair State University**, Montclair, NJ
B.S. in Chemistry, ACS Certified

Teaching and Mentoring Experience

Aug 2024 – Present | **Assistant Professor of Chemistry, Trine University**

- Teach undergraduate lecture and laboratory courses in general chemistry, physical chemistry, and data science, including CH 104, CH 114, CH 114L, CH 352, CH 354, CH 354L, CH 374, and CH 374L.
- Created CH 352: Data Science in Chemistry and Biology; redesigned CH 104: General Chemistry I around an atoms-first approach; and revised CH 114L: General Chemistry II Laboratory to emphasize computer skills and quantitative data analysis.
- Mentor undergraduate students in computational chemistry, cheminformatics, bioinformatics, Python programming, and high-throughput/high-performance computing workflows.
- Advise undergraduate students on academic planning, course selection, and career preparation.

Sep 2021 – May 2022 | **Navari Family Center for Digital Scholarship Fellow, University of Notre Dame**

- Designed and facilitated $\text{\LaTeX}$ workshops for students, faculty, and staff, including introductory and thesis/dissertation writing workshops.

Sep 2017 – July 2019 | **Graduate Teaching Assistant, University of Notre Dame**

- Supported instruction in General Chemistry and Physical Chemistry through problem-solving sessions, office hours, grading, and student feedback.

Sep 2015 – May 2017	Graduate Teaching Assistant, Montclair State University Supported Principles of Chemistry Laboratory for non-science majors through laboratory preparation, grading, and instructional support.
---------------------	--

Research Experience

Jan 2023 – July 2024	Postdoctoral Fellow, University of Notre Dame <i>Computational Catalytic Chemist</i> - Applied density functional theory and molecular dynamics simulations to investigate structure-property relationships in zeolite catalysis, including Al-density-dependent dealumination in CHA zeolites. - Collaborated with experimental researchers, mentored graduate and undergraduate researchers, coordinated group meetings, and presented findings at national conferences.
Jan 2018 – Dec 2022	Graduate Research Assistant, University of Notre Dame <i>Computational Catalytic Chemist</i> - Used density functional theory, molecular dynamics, and statistical mechanical models to study catalytic reactions on surfaces, in plasmas, and in zeolite materials. - Investigated free energy and entropy models for surface-bound species, guest-host interactions in zeolite crystallization, Al distributions in zeolites, and nitrogen-containing plasma synthesis intermediates. - Developed and delivered Python and $\text{\LaTeX}$ workshops to support scientific computing, reproducible research, and technical writing.
Jan 2014 – July 2017	Graduate Research Assistant, Montclair State University <i>Computational Chemist</i> - Performed computational studies of C–CN bond activation through nickel catalysis using electronic structure methods, including random phase approximation approaches. - Developed computational protocols for evaluating reaction pathways, energetics, and mechanistic hypotheses.

Research Interests

Computational chemistry; density functional theory; molecular modeling; cheminformatics; machine learning for molecular property prediction; scientific computing education; undergraduate research in chemistry, biology, coding, and data science.

Grants

Jan 2026	Equipment: Enhancing Scientific Computing Infrastructure for Undergraduate Chemistry and Biology Education National Science Foundation, NSF 23-510. Principal Investigator. Amount requested: \$263,953. Status: Pending.
----------	---

Publications

- | | | |
|------|----|--|
| 2023 | 1. | Waitt, C. , Gao, X., Gounder, R., Debellis, A., Prasad, S., Moini, A. & Schneider, W. F. Analysis and Augmentation of Guest–Host Interaction Energy Models as CHA and AEI Zeolite Crystallization Phase Predictors. <i>J. Phys. Chem. C</i> 127 , 22740–22741 (2023). |
| 2022 | 2. | Yan, C., Waitt, C. , Akintola, I., Lee, G., Easa, J., Clarke, R., Geng, F., Poirier, D., Otor, H. O., Rivera-Castro, G., Go, D. B., O’Brien, C. P., Hicks, J. C., Schneider, W. F. & Ma, H. Recent Advances in Plasma Catalysis. <i>Industrial & Engineering Chemistry Research</i> 61 , 7675–7678 (2022). |
| 2021 | 3. | Barboun, P. M., Daemen, L. L., Waitt, C. , Wu, Z., Schneider, W. F. & Hicks, J. C. Inelastic Neutron Scattering Observation of Plasma-Promoted Nitrogen Reduction Intermediates on Ni/ γ -Al ₂ O ₃ . <i>ACS Energy Letters</i> 6 , 2048–2053 (2021). |
| | 4. | Waitt, C. , Miles, R. A. & Schneider, W. F. Adsorbate Free Energies from DFT-Derived Translational Energy Landscapes. <i>J. Phys. Chem. C</i> 125 , 20331–20342 (2021). |
| 2017 | 5. | Tummalapalli, S. R., Bhat, R., Waitt, C. , Eshuis, H. & Rotella, D. P. Synthesis and computational analysis of conformationally restricted [3.2. 2]- and [3.2. 1]-3-azabicyclic diamines. <i>Tetrahedron Letters</i> 58 , 4087–4089 (2017). |
| 2016 | 6. | Waitt, C. , Ferrara, N. M. & Eshuis, H. Thermochemistry and geometries for transition-metal chemistry from the random phase approximation. <i>Journal of Chemical Theory and Computation</i> 12 , 5350–5360 (2016). |

Presentations

- | | |
|-------------|---|
| March 2023 | Analysis and Augmentation of Guest-Host Interaction Energy Models as CHA and AEI Zeolite Crystallization Phase Predictors
Oral presentation, ACS Spring 2023, American Chemical Society. |
| August 2022 | Exploring the Relationship Between Organic Structure Directing Agents and Al Distribution in CHA
Oral presentation, ACS Fall 2022, American Chemical Society. |
| May 2022 | Exploring the Relationship Between Organic Structure Directing Agents and Al Distribution in CHA
Poster presentation, NAM 27, North American Catalysis Society. |
| April 2021 | Moving Beyond the Harmonic Oscillator and Hindered Translator Models to Compute Free Energies and Entropies of Surface-Bound Species
Oral presentation, ACS Spring 2021, American Chemical Society. |

Service

Aug 2025 – Present	Educational Resources Committee, Trine University • Serve as committee member.
Aug 2025 – Present	AI Ad Hoc Committee, Trine University • Serve as committee member.
Aug 2024 – Present	Departmental and University Service, Trine University • Serve as an outreach volunteer, Scholarship Day participant, and undergraduate academic advisor.

Awards, Fellowships, and Honors

Sep 2021	Navari Family Center for Digital Scholarship Pedagogy Fellowship
May 2019	Rudy Bottei Graduate Teaching Award
May 2016	Margaret and Herman Graduate Summer Research Fellowship
Sep 2010	Margaret and Herman Freshman Scholarship

Professional Development

July 2026	Selected Participant, OSG/HTC School; week-long training in high-throughput computing, distributed computing workflows, and scalable research computing for computational research.
-----------	---

Technical Skills

Computational Chemistry: Density functional theory, molecular dynamics, molecular modeling, thermochemistry, free energy models

Scientific Computing and Data Science: Python, NumPy, RDKit, machine learning workflows, cheminformatics, high-performance computing, high-throughput computing

Software and Tools: VASP, Turbomole, Quantum ESPRESSO, ORCA, ASE, Git, $\text{\LaTeX}$